

CODERUSH 2026 • ML MODULE

Economic Class Classification

Participant Rulebook & Dataset Guide

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1. Problem Overview

📖 Problem Statement

Given a dataset where each sample consists of multiple individuals described by their demographic and economic attributes, predict the **economic class label** of the entire sample. Each sample is identified by a **bag_id** and may contain a variable number of individuals.

This is a **Multiclass Classification** problem. You are not predicting a label for each individual — you are predicting a single label for the *entire group* (bag) based on the collective attributes of all its members.

1.1. Target Labels

Each **bag_id** must be assigned exactly one of the following three economic class labels:

Label	Class	Description
0	lower	Individuals in the lower economic tier
1	middle	Individuals in the middle economic tier
2	upper	Individuals in the upper economic tier

2. Input Data Format

2.1. Bag Structure

Each sample in the dataset is a *bag* — a collection of rows where each row represents one individual. Key structural properties:

- Every sample is identified by a unique **bag_id**.
- A bag contains a **variable number of individuals** (rows).
- Individuals within a bag are **unordered** — the row index carries no semantic meaning.
- Each bag should be treated as a **set of instances**.

2.2. Row Structure

Each row in the dataset represents one individual and contains the following fields:

Column	Description
bag_id	Unique identifier for the group
bag_size	Number of individuals in the group
person_idx	Index of the individual within the group
annual_hours_est	Estimated total working hours per year based on weekly commitment
capital_activity_flag	Indicator showing if any capital gain or loss was reported
capital_gain	Income gained from capital investments
capital_loss	Income lost from capital investments
currency_code	Currency unit used for financial columns
education	Highest education qualification attained
education_num	Numerical representation of education level
education_tier	Administrative grouping of educational attainment levels
hours_per_week	Number of hours the individual typically works per week
interviewer_id	Unique identifier of the administrator who conducted the survey
interview_mode	Mode of data collection used during the survey
is_adult_flag	Indicator for individuals aged 18 and older
marital_status	Current marital status of the individual
native_country	Country of origin
net_capital_asset	Calculated difference between capital gains and losses
occupation	Job or profession of the individual
poverty_line_usd	Reference poverty threshold value used in the dataset
processing_flag	Data quality indicator flag
race	Self-identified race category
relationship	Relationship role within the original survey context
sex	Gender of the individual
survey_duration_mins	Time taken (in minutes) to complete the survey interview
survey_year	Year in which the data was collected
workclass	Type of employment or work sector
year_of_birth	Birth year of the individual
label	Target label (Lower, Middle, or Upper) — <i>only in training data</i>

3. Dataset Files

The following files are provided for this competition:

File	Split	Contents
Coderush-26-ML-Train.csv	Training	All features + labels (repeated per row within each <code>bag_id</code>)
Coderush-26-ML-test.csv	Test	Features only — used for Kaggle leaderboard submissions

✖ Important

Labels are **only available in the training set**. The test file has no label column. The judges retain a separate **held-out test set** used exclusively for final evaluation — it is never released to participants.

4. Submission System

✖ DISQUALIFICATION NOTICE

Read this section very carefully. Adherence to these submission guidelines is **strictly mandatory**. Failure to adhere to these guidelines will result in the **PERMANENT REMOVAL** of your entry from the competition.

Zero appeals will be entertained after the 08:00 AM deadline.

In addition to your `submission.csv`, you **must** share your kaggle notebook with the Organizers on Kaggle and give the edit access so that we can run your model on the hidden test set.

- Your `submission.csv` on the test set that should be reproducible by running your notebook.
- Your saved model checkpoint file.

4.1. Required Notebook Outputs

When you click “Save Version” (Save & Run All) on Kaggle, your notebook must generate and save the following two items to the output directory:

4.1.1. Submitted CSV

Your code must output a CSV file containing predictions for every `bag_id` in the validation set.

```
bag_id,label
10001,0
10002,2
10003,1
10004,0
...
```

- The file must contain **exactly two columns**: `bag_id` and `label`.
- `label` values must be mapped to numbers: **0** (lower), **1** (middle), or **2** (upper).
- **Every** `bag_id` from the validation set must be present.
- No extra columns, no missing rows.

4.1.2. Saved Model Files

Your script must save your best trained model file(s) to the Kaggle working directory so it appears in the notebook's Output tab. This ensures your model is exactly 1-to-1 reproducible for verification purposes.

Accepted model formats:

- Scikit-learn / joblib: `.pkl`, `.joblib`
- PyTorch: `.pt`, `.pth`
- Keras / TensorFlow: `.h5`
- XGBoost / LightGBM native formats

4.2. Code Requirements

★ Reproducibility & Hidden Test Set Evaluation

- **Runnable Code:** The notebook must be runnable end-to-end without crashing. Include clear cells for data loading, preprocessing, training, and evaluation.
- **Hidden Test Set:** Your data preprocessing and inference code must be dynamic. During final evaluation, judges will run your saved notebook and model on a **Secret Hidden Test Set** to verify robustness and ensure no hard-coding or data leakage occurred.

4.3. Organizers' Kaggle IDs:

- rehanfarooqs
- muhammadshafaybutt

5. Deadlines & Competition Rules

5.1. Key Dates

Event	Date	Time (PKT)
Submission Deadline (Kaggle)	Saturday, 9 May 2026	08:00 AM
Presentation Day	Sunday, 10 May 2026	TBA

5.2. Strict Deadline Policy

✗ Zero Tolerance After Deadline

Once the deadline passes:

- Kaggle submissions will be **closed**.
- **No late submissions** of any kind will be accepted.
- **No notebook changes**, model updates, or resubmissions are permitted through any channel (including direct messages to organisers).
- **Latest notebook version** before the deadline would be considered
- **All submissions are final**.

Only work submitted before the deadline will be evaluated on Presentation Day.

5.3. Presentation Day

On **Sunday, 10 May 2026**, shortlisted participants will present:

- Their submitted notebook
- The best model — its architecture, training choices, and results
- Their data understanding and modelling rationale
- Visualisations and insights from exploratory data analysis

📌 Presentation Tip

Judges reward **clarity and reasoning** over complexity. A well-explained simple model will score higher than a poorly-explained complex one.

6. Evaluation & Judging Criteria

Final ranking is determined by a combination of leaderboard performance and qualitative judging on Presentation Day.

6.1. Leaderboard Evaluation (Kaggle)

Metric	Role	Description
Macro F1 Score	Primary	Unweighted mean of per-class F1 scores across all three labels
Middle Class F1	Tie-breaker	F1 score specifically for the middle class

Macro F1 is defined as:

$$F1_{\text{macro}} = \frac{1}{3} \sum_{c \in \{\text{lower, middle, upper}\}} F1_c \quad \text{where} \quad F1_c = \frac{2 \cdot P_c \cdot R_c}{P_c + R_c}$$

🔍 Why Macro F1?

Macro F1 weights all three classes equally regardless of their frequency in the dataset. This means models that ignore minority classes (e.g. always predict **middle**) will be penalised. You must perform well across *all three* classes.

6.2. Suggested Scoring Weights

The final score is a weighted combination of leaderboard performance and qualitative criteria evaluated on Presentation Day:

#	Criterion	Evaluated By
1	Leaderboard Performance (Macro F1)	Kaggle
2	Simplicity & Approach Preference	Judges
3	Pipeline Quality & Reproducibility	Judges
4	Data Understanding & EDA	Judges
5	Visualisation & Communication	Judges

6.3. Simplicity & Approach Preference

This is a **core judging principle**: simpler approaches are rewarded more heavily when performance is comparable. Priority goes from simple classical algorithms, to ensemble and tree-based models, to shallow networks, to deep learning — in increasing order of complexity. The performance gap must be **substantial** to justify moving up the complexity ladder.

✓ Scoring Principle

A simple model achieving 0.70 Macro F1 will rank **above** a deep model achieving 0.72 Macro F1, as the performance difference is marginal and the simplicity advantage is significant.

6.4. Penalty for Hacks

✗ Penalty for Hacks

Any attempt to game the evaluation — including hard-coding predictions, reverse-engineering validation labels, or exploiting knowledge of the test set in any form — will be caught during held-out test set evaluation. Such submissions will receive a **heavy score penalty** and may result in **immediate disqualification** from the competition. All submitted notebooks and models are subject to review.

7. Summary Checklist

Use this checklist before the deadline:

- ☐ Predictions submitted to Kaggle in correct CSV format (`bag_id`, `label`)
- ☐ All `bag_ids` from the test set included
- ☐ Labels are lowercase numeric (0, 1, 2)
- ☐ Notebook runs end-to-end without errors
- ☐ Best model files saved in the kaggle notebook
- ☐ Kaggle notebook **shared with the organizers**, see [4.3](#)
- ☐ All submissions completed before **9 May 2026, 08:00 AM PKT**

8. Quick Reference

Item	Detail
Competition Name	Coderush 2026 — ML Module
Problem Name	Economic Class Classification
Dataset Name	Coderush 2026 ML Dataset
Task Type	Multi-class classification (bag-level)
Classes	lower, middle, upper
Primary Metric	Macro F1 Score
Tie-Breaker Metric	Middle Class F1
Submission Platform	Kaggle
Deadline	Saturday, 9 May 2026 — 08:00 AM PKT
Presentation Day	Sunday, 10 May 2026

Coderush 2026 — ML Module • Economic Class Classification

This document is the official participant guide. All rules are final.